

BT101 - Ecology, Biodiversity & Evolution-I

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What is ecological niche.

An ecological niche describes how a species lives and interacts with other organisms in a habitat. It can be thought of as the role or job that a species has in nature.

Characters of Bacteria:

Bacteria play many roles in our ecosystem. Bacteria are decomposers which break down dead material and recycle it. By decomposing dead organic matter, they contribute to soil structure and the cycles of nature. Bacteria provide nutrients like nitrogen and phosphorus to plants as well as to produce growth hormones. They are also needed for the carbon and nitrogen cycles.

Layers of atmosphere and their heights:

- ❖ Exosphere: 700 to 10,000 km (440 to 6,200 miles)
- ❖ Thermosphere: 80 to 700 km (50 to 440 miles)
- ❖ Mesosphere: 50 to 80 km (31 to 50 miles)
- ❖ Stratosphere: 12 to 50 km (7 to 31 miles)
- ❖ Troposphere: 0 to 12 km (0 to 7 miles)

What is limits selection:

1= Evolution requires genetic variation 2= Gene interactions affect fitness of alleles 3= Genes have multiple effects.

Structure and Function of Golgi Apparatus.

Structure:

It is Located near the end of the ER close to the nucleus and Composed of several layers of cisternae (fluid-filled sacs) • It Has three different parts: 1. cis-Golgi 2. medial-Golgi 3. trans-Golgi

Functions:

1= cis is first cisternae. 2= trans is final cisternae. 3= From trans, proteins are packaged into vesicles destined to lysosomes, secretory vesicles, or the cell surface. Transport vesicles are used

to move back and forth between the ER and Golgi bodies 4= One side of the Golgi body receives transport vesicles produced by the ER. 5= This side is called cis face. 6= Opposite side releasing vesicle is trans face.

What is the difference between Analogy & homology?

Homology:

Homology involves the study of organs that have evolved from same origin or from common ancestor but have different functions. It is a Divergent evolution.

For example:

- ❖ Wings of bat and Hands of humans.

Analogy:

Analogy involves the study of organs that are evolved from different ancestors but perform a similar function. It is a Convergent Evolution.

For example:

- ❖ Wings of bat and wings of insects.

<u>Homology</u>	<u>Analogy</u>
In biology, homology is the resemblance of the arrangement, physiology, or growth of various species of organisms.	In biology, an analogy is a functional similarity of structure, based on the similarity of use and not upon common evolutionary origins.
Due to a similar structure, they do have similar functions	Due to different structures, they do not have similar functions.
Homologous structures are inherited from a common precursor.	Analogous structures are inherited from a different precursor.
Homologous structures are developed from related embryonic substances.	Homology is developed in species that are not related to each other.
Homology happens due to divergent evolution.	Analogy happens due to convergent evolution

Flow of Energy in Ecosystem Feeding Levels: or Explain Feeding level of organisms:

- ❖ Food energy passes through different sequence.
- ❖ Example: plants → grass hopper → frog → snake → hawk
- ❖ Many plant species eaten by insects
- ❖ Many insects species eaten by frogs
- ❖ this complexity is organized by ecologists.
- ❖ Recognized limited number of feeding or trophic levels
- ❖ In Food Chain
- ❖ Food energy passes through different sequence.
- ❖ Level 1: Primary producer
- ❖ Level 2: Primary consumer
- ❖ Level 3: Secondary consumer
- ❖ Level 4: Tertiary Consumer Level
- ❖ 5: Apex Predator

Reproduction in arthropods?

Arthropods reproduce by sexual reproduction, which involves the generation and fusion of gametes. Most arthropods are either male or female, and they undergo internal fertilization. Once

the egg has been fertilized, the female usually lays the egg, and it continues developing outside of the mother’s body.

Reproductive System:

1= Mostly dioecious 2= Paired genital opening 3= Male often packages sperm in a spermatophore 4= Transfer them into the female genital tract 5= Copulation occurs. 6= Sperms are transferred via modified pedipalp.

What is autolysis and putrefaction:

Autolysis: Breaking down of tissues by the body's own internal chemicals and enzymes

Putrefaction: Breakdown of tissues by bacteria. And releases compounds such as cadaverine and putrescine causing decaying odor.

Write a note on rate of evolution:

The rate of evolution is quantified as the speed of genetic or morphological change in a lineage over a period of time. The speed at which a molecular entity (such as a protein, gene, etc.) evolves is of considerable interest in evolutionary biology.

Carbon cycle:

- ❖ Carbon enters the atmosphere as CO2.
- ❖ CO2 is absorbed by autotrophs such as green plants.
- ❖ Animals consume plants, thereby, incorporating carbon into their system.
- ❖ Animals and plants die, their bodies decompose and carbon is reabsorbed back into the atmosphere.

Exponential growth and Logistic growth.

Exponential growth	Logistic growth
It occurs when resources are abundant	It occurs when resources are limited
Population passes well beyond the carrying capacity of the ecosystem	Population doesn’t grow beyond the carrying capacity of the ecosystem
It produce J shaped curve	It produce S shaped curve
A stationary or stable phase is completely achieved	A stationary or stable phase is achieved
It has two phases lag and log	It has four phases lag, log, deceleration and steady
It occurs in fewer organisms like lemmings etc.	It is more common in wildlife, yeast etc.
$DN/dt=Nr$ N= population size, r= intrinsic rate	$DN/dt= rN(k-N)/K$ $DN/dt=$ change in population size per unit time K= carrying capacity

Characteristics of Archaea:

The common characteristics of Archaeobacteria known to date are these: 1= the presence of characteristic tRNAs and ribosomal RNAs; 2= the absence of peptidoglycan cell walls, with in many cases, replacement by a largely proteinaceous coat; 3= the occurrence of ether linked lipids built from phytanyl chains. 4= Most don’t need oxygen to survive 5= They can produce ATP from sunlight 6= They can survive enormous temperature and pressure extremes. 7= They can survive high doses of radiation 8= They can survive under rocks and in ocean floor vents deep below the ocean’s surface.

Characteristics of Mammalian:

- ❖ Female mammals have mammary glands. The glands produce milk after the birth of offspring. Milk is a nutritious fluid. It contains disease-fighting molecules as well as all the nutrients a baby mammal needs. Producing milk for offspring is called lactation.
- ❖ The skin of many mammals is covered with sweat glands. The glands produce sweat, the salty fluid that helps cool the body.
- ❖ Mammalian lungs have millions of tiny air sacs called alveoli. They provide a very large surface area for gas exchange.
- ❖ The heart of a mammal consists of four chambers. This makes it more efficient and powerful for delivering oxygenated blood to tissues.
- ❖ The brain of a mammal is relatively large and has a covering called the neocortex. This structure plays an important role in many complex brain functions.
- ❖ The mammalian middle ear has three tiny bones that carry sound vibrations from the outer to inner ear. The bones give mammals exceptionally good hearing. In other vertebrates, the three bones are part of the jaw and not involved in hearing.
- ❖ Mammals have four different types of teeth. The teeth of other vertebrates, in contrast, are all alike.

Extinction and its causes:

Termination of an organism or of a group of organisms (taxon), usually a species. Moment of extinction is death of the last individual of the species. Capacity to breed and recover may have been lost before this point.

Variety of causes, independently or in concert, are responsible for extinctions. • Disruption of habitat • Loss of ecosystem interactions e.g. disrupted food chain or web, symbiosis, predation • Pollution • Loss of genetic variation • Catastrophic disturbances • Natural or human-caused activities Habitat loss is the most important cause of modern-day extinction. Human activities are negatively affecting all types of habitat, from rain forest to ocean floor.

Natural habitats may be adversely affected by humans in three ways.

1. Habitat destruction 2. Pollution 3. Habitat fragmentation

Phylum plathelminthes and classifications ?

Platyhelminthes are triploblastic, bilaterally symmetrical, dorsoventrally flattened, acoelomate flatworms with organ grade of construction without a definite anus, circulatory, skeletal or respiratory system but with Protonephridial excretory system and mesenchyme filling the space between the various organ of the body.

Classification:

- 1) Class Turbellaria (Free-Living Flatworms)
- 2) Class Monogenea
- 3) Class Trematoda (Flukes)
- 4) Class Cestoidea (Tapeworms)

Types of omnivores:

An omnivore is a kind of animal that eats either other animals or plants. Some omnivores will hunt and eat their food, like carnivores, eating herbivores and other omnivores. Some others are scavengers and will eat dead matter. Many will eat eggs from other animals. Omnivores generally occupy the third trophic level alongside meat-eating carnivores. Omnivores are a diverse group of animals.

Examples of omnivores include bears, birds, dogs, raccoons, foxes, certain insects, and even humans.

6 Classes of arthropods:

- ❖ Arachnid: the Class of Arthropods that includes spiders, mites, ticks, scorpions, pseudoscorpions and harvestmen.
- ❖ Chilopoda: the Chilopoda are a group of predatory arthropods more commonly known as centipedes.
- ❖ Collembola: the Collembola/springtails are one of the three non-insect hexapod classes of invertebrates.
- ❖ Crustaceans: a large group within of the Arthropods including crabs, shrimps and lobsters.
- ❖ Diplopoda: the Diplopoda are a group of arthropods more commonly known as millipedes.
- ❖ Diplura: the Diplura/two-pronged bristletails are one of the three non-insect hexapod classes of invertebrates.
- ❖ Insects: the largest Class of arthropods and the most diverse group of animals in the world.
- ❖ Myriapoda: a term previously used to describe the group of arthropods containing the millipedes and centipedes.
- ❖ Non-insect hexapods: the grouping of three classes of six-legged invertebrates that are not insects.
- ❖ Protura: the Protura are one of the three non-insect hexapod classes of invertebrates.

Binomial nomenclature?

The system of nomenclature in which two terms are used to denote a species of living organism, the first one indicating the genus and the second the specific epithet. The binomial system of nomenclature brings order to a chaotic world of common names. Common names vary from country to country, and from region to region within a country. There was need of globally recognized names.

In binomial system, no two kinds of animals have the same binomial name. Every scientific name has two parts:

1. First from genus 2. Second from species Names are written in italics

Define transpiration, ground water and percolation?

Transpiration: Plants absorb water from the soil. Water moves from the roots through the stems to the leaves. The process of evaporation through plant leaves is called transpiration.

Groundwater: is the water found underground in the cracks and spaces in soil, sand and rock. It is stored in and moves slowly through geologic formations of soil, sand and rocks called aquifers.

Percolation (Infiltration): Percolation is an important process. Rain water soaks into (infiltrates) the ground, into the soil and underlying rock layers. Some of this water ultimately returns to the surface at springs or in low spots downhill. Some of the water percolates underground It is called groundwater.

What are the diagnostic features of phylum Mollusca?

1= Body divided into three parts 2= Head-foot, visceral mass and mantle 3= Mantle secretes a calcareous shell 4= Bilateral symmetry 5= Coelom reduced to cavities surrounding the heart, nephridia and gonads.

Difference between grazing and browsing ?

1= Grazers Feeds on plants such as grasses, soft flowering plants (forbs) and algae. Plants are not killed. Examples: Deer & Snails

Types of Grazing:

1. Graminivory Feeding primarily on grass. Specifically "true" grasses in family Poaceae. Examples: Horses, cattle.

2. Coprophagy Feeding on grass, forbs, leafy weeds, fruits, tree bark Heavy grazing for half an hour of grazing period Followed by half an hour of selective feeding. Example: Rabbit

2= Browsers Feed on non-grass plant material Leaves, soft shoots, twigs, fruits, woody plants such as shrubs Examples: Goats, deer, giraffe, elephants.

Characteristics of colentrata:

- ❖ These are mostly aquatic or marine habitat animals.
- ❖ These species exhibit a tissue-level organization.
- ❖ The mouth is enclosed by thin and short tentacles.
- ❖ They are diploblastic animals, in which, the body is made up of two layers of cells:
- ❖ Ectoderm – One layer makes up the cells outside the body
- ❖ Endoderm – the other forms the inner lining of the body.
- ❖ They have cavities in their body.
- ❖ The body is radially symmetrical.
- ❖ The digestion is both intracellular and extracellular.
- ❖ The nervous system and the circulatory system is absent.
- ❖ They excrete and respire through simple diffusion.
- ❖ The mode of reproduction is asexual, which is through budding.
- ❖ The sexual mode of reproduction is seen only in a few Coelenterates. Hydra is the example.

What is highly conserved part in birds:

All birds are easily recognizable as they because of highly conserved structures

1. Feathers 2. Bills 3. Body form

Particular parts of birds, however, are less conservative and have a higher rate of change.

Metamorphosis and its types:

Metamorphosis is a process by which animals undergo extreme, rapid physical changes some time after birth. The result of metamorphosis may be change to the organism's entire body plan, such as a change in the animal's number of legs, its means of eating, or its means of breathing.

Types of Metamorphosis:

1= Complete Metamorphosis: In complete metamorphosis, a larva completely changes its body plan to become an adult. The most famous example is that of the butterfly, which starts out as a worm-like, leaf-eating caterpillar and transforms into a flying, nectar-drinking creature with an exoskeleton.

2= Incomplete Metamorphosis: In incomplete metamorphosis, only some parts of the animal's body change during metamorphosis. Animals that only partially change their bodies as they mature are called "hemimetabolous."

Oscillating and frequency selection?

Oscillating Selection: In some cases, selection favors one phenotype at one time and another phenotype at another time. Example: Medium ground finches of the Galápagos Islands.

Frequency dependent selection: This type of selection favors certain phenotypes depending on how commonly or uncommonly they occur. It has two types: 1. Negative 2. Positive

Examples of zoomastigophoras?

Trypanosoma brucei: It has three sub species.

1= T. b. brucei: Parasite of non human. 2= T. b. gambiense: Causes sleeping sickness in humans
3= T. b. rhodesiense: Causes sleeping sickness in humans.

Effects of Global Warming:

Following are the major effects of global warming:

Rise in Temperature: Global warming has led to an incredible increase in earth's temperature. Since 1880, the earth's temperature has increased by ~1 degrees. This has resulted in an increase in the melting of glaciers, which have led to an increase in the sea level. This could have devastating effects on coastal regions.

Threat to the Ecosystem: Global warming has affected the coral reefs that can lead to a loss of plant and animal lives. Increase in global temperatures has made the fragility of coral reefs even worse.

Climate Change: Global warming has led to a change in climatic conditions. There are droughts at some places and floods at some. This climatic imbalance is the result of global warming.

Spread of Diseases: Global warming leads to a change in the patterns of heat and humidity. This has led to the movement of mosquitoes that carry and spread diseases.

Decrease in the Human Population: Due to an increase in floods, tsunamis and other natural calamities, the population of humans tend to decrease. Also, the spread of diseases leads to a decrease in the human population.

Loss of Natural Habitat: A global shift in the climate leads to the loss of habitats of several plants and animals. In this case, the animals need to migrate from their natural habitat and many of them even get extinct.

Write a note on Aposematism:

Aposematism (Greek, apo = away, sematic = sign) is the use of warning coloration to inform potential predators that an animal is poisonous, venomous, or otherwise dangerous. Oftentimes orange or red patterns may be warnings (as in coral snakes), but do not assume that red is always a warning. It is the advertising by an animal to potential predators that it is not worth attacking or eating. This unprofitability may consist of any defences which make the prey difficult to kill and eat, such as toxicity, venom, foul taste or smell, sharp spines, or aggressive nature.

Food chain:

A food chain is a linear network of links in a food web starting from producer organisms and ending at an apex predator species, detritivores, or decomposer species. A food chain also shows how organisms are related to each other by the food they eat. Each level of a food chain represents a different trophic level.

What is the meaning by aestivation?

Aestivation is typically defined as a summer or dry season dormancy. The organisms respond to unfavorable conditions by suspending their activities and go into a period of deep sleep. Summer sleep is called aestivation.

Waste water pollution:

Water pollution occurs when harmful substances—often chemicals or microorganisms—contaminate a stream, river, lake, ocean, aquifer, or other body of water, degrading water quality and rendering it toxic to humans or the environment. This widespread problem of water pollution is jeopardizing our health.

What is difference between monocots and dicots.

Monocots develop a fibrous root system where no single root serves as the origin for all of the other roots. Dicots develop a taproot system where a single root descends from the embryo (seed) and branches out into a network of smaller roots and finally root hairs that absorb water and nutrients.

Write a note on Origin of eukaryote?

The leading hypothesis, called the endosymbiotic theory, is that eukaryotes arose as a result of a fusion of Archaean cells with bacteria, where an ancient Archaean engulfed (but did not eat) an ancient, aerobic bacterial cell.

Different types of flagella.

1. Monotrichous; one flagella 2. Lophotrichous; tuft at one end 3. Amphitrichous; both ends 4. Peritrichous; all around bacteria.

Define atmosphere.

The atmosphere is a mixture of gases that surrounds the Earth. It helps make life possible by providing us with air to breathe, shielding us from harmful ultraviolet (UV) radiation coming from the Sun, trapping heat to warm the planet, and preventing extreme temperature differences between day and night.

What are the Autotrophs:

An autotroph is an organism that can produce its own food using light, water, carbon dioxide, or other chemicals. Because autotrophs produce their own food, they are sometimes called producers. Plants are the most familiar type of autotroph, but there are many different kinds of autotrophic organisms.

Types: 1= Photoautotrophs, 2= Chemoautotrophs

1= Photoautotrophs: a photosynthetic organism (such as a green plant or a cyanobacterium) that utilizes energy from light to synthesize organic molecules. Examples: plants, algae, and cyanobacteria

2= Chemoautotrophs: Chemoautotrophs occur in dark habitats. They obtain energy by means of inorganic oxidation reactions. Convert toxic molecules into glucose Example: Microbes, mussels.

Characteristics of coelomate:

Coelomate animals or Coelomata (also known as eucoelomates – "true coelom") have a body cavity called a coelom with a complete lining called peritoneum derived from mesoderm (one of the three primary tissue layers). The complete mesoderm lining allows organs to be attached to each other so that they can be suspended in a particular order while still being able to move freely within the cavity. Most bilateral animals, including all the vertebrates, are coelomates.

Theory of natural selection:

Natural selection is the process through which populations of living organisms adapt and change. Through this process of natural selection, favorable traits are transmitted through generations. Natural selection can lead to speciation, where one species gives rise to a new and distinctly different species.

Digestive system of echinoderms:

Echinoderms have a simple digestive system with a mouth, stomachs, intestine and anus. In many, the mouth is on the underside and the anus on the top surface of the animal. Sea stars can push their stomachs outside of their body and insert it into its prey allowing them to digest the food externally.

Evidence of evolution of mitochondria:

Mitochondria originated by permanent enslavement of purple non-sulphur bacteria. These endosymbionts became organelles through the origin of complex protein-import machinery and insertion into their inner membranes of protein carriers for extracting energy for the host.

Evidence of the natural selection?

Darwin based his theory on many solid evidences from his observations and collected samples.

Three major evidences:

- ❖ Geology
- ❖ Fossil Evidences
- ❖ Galápagos islands

Types of symbiosis”

1. Mutualism 2. Commensalism 3. Parasitism

Advantages of atmosphere:

The atmosphere is incredibly important. It protects us from harmful solar radiation (such as in the case of the ozone layer). It circulates dust, gases and clouds. The oxygen-rich lower atmosphere is good for life. It is also important for birds and other flying creatures. It regulates the temperature.

Ozone Depleting Substances:

Halocarbon refrigerants , Solvents , Propellants in aerosols, Chlorofluorocarbons, Hydro chlorofluorocarbons, Methyl bromide, Carbon tetrachloride, Methyl chloroform.

Autotrophs and heterotrophs difference:

Autotrophs are known as producers because they are able to make their own food from raw materials and energy. Examples include plants, algae, and some types of bacteria. Heterotrophs are known as consumers because they consume producers or other consumers. Dogs, birds, fish, and humans are all examples of heterotrophs.

Characteristics of r-selected species:

1= High growth rates 2= Typically exploit less crowded ecological habitats 3= Produce many offspring 4= Relatively low probability of surviving to adulthood 5= High reproductive rate supports their survival. 6= Especially in unstable environment.

Characteristics of Phylum actinopoda:

1= Mostly marine 2= Spherical, planktonic 3= Axopodia supported by microtubules 4= Includes marine radiolarians with siliceous tests 5= About 3,000 species.

Note on decomposition:

Decomposition is the process by which dead organic substances are broken down into simpler organic or inorganic matter such as carbon dioxide, water, simple sugars and mineral salts. The process is a part of the nutrient cycle and is essential for recycling the finite matter that occupies physical space in the biosphere. Bodies of living organisms begin to decompose shortly after death. Animals, such as worms, also help decompose the organic materials. Organisms that do this are known as decomposers. Although no two organisms decompose in the same way, they all undergo the same sequential stages of decomposition. The science which studies decomposition is generally referred to as taphonomy from the Greek word taphos, meaning tomb. Decomposition can also be a gradual process for organisms that have extended periods of dormancy.

Describe evolution and paleontology?

Paleontology is the study of fossils. It provides some of the most direct evidence for evolution. Fossils (to dig) are evidence of plants and animals that existed in the past and have become incorporated into the earth's crust.

Evolution is change in the heritable characteristics of biological populations over successive generations.

Regeneration in echinoderms:

All echinoderms have the capability to regenerate some or most organs or appendages. Since the regenerated structures are usually identical to the lost ones, including limbs, disc, gut, spines and podia and, in some species, regeneration is used for asexual reproduction. It has then been assumed that formation of new muscle cells or myogenesis occurs in all echinoderms.

Vestigial Organs:

Vestigial organs are rudimentary anatomical structures that are retained in a species despite having lost their primary ancestral function. These structures often lack an apparent purpose, in contrast to the full functionality of these organs observed in closely related and ancestral species.

What is cypsis explain with reference to skin color and counter shading

The ability of an organism to conceal itself especially from a predator by having a color, pattern, and shape that allows it to blend into the surrounding environment.

Camouflage, also called cryptic coloration, is a defense or tactic that organisms use to disguise their appearance, usually to blend in with their surroundings. Organisms use camouflage to mask their location, identity, and movement. This allows prey to avoid predators, and for predators to sneak up on prey.

Types of polluted water:

Contaminants such as chemicals, nutrients, and heavy metals are carried from farms, factories, and cities by streams and rivers into our bays and estuaries; from there they travel out to sea. Meanwhile, marine debris—particularly plastic—is blown in by the wind or washed in via storm drains and sewers.

What is endomembrane system?

The endomembrane system (endo- = “within”) is a group of membranes and organelles in eukaryotic cells that works together to modify, package, and transport lipids and proteins.

The endomembrane system is composed of the different membranes that are suspended in the cytoplasm within a eukaryotic cell. These membranes divide the cell into functional and structural compartments, or organelles.

Rate of evolution:

The rate of evolution is a measurement of the change in an evolutionary lineage over time. Three factors that determines the evolution rate are: 1= Mutation rate 2= Probability of new mutation 3= Population size

Define evolution:

Evolution is the change in the characteristics of a species over several generations and relies on the process of natural selection.

Limiting factor effect the size of population:

Limiting factors are resources or other factors in the environment that can lower the population growth rate. Limiting factors include a low food supply and lack of space. Limiting factors can lower birth rates, increase death rates, or lead to emigration.

Limiting factors regulate population size:

- ❖ Rate of Birth
- ❖ Rate of Death/mortality
- ❖ Dispersal
- ❖ Food Resources
- ❖ Space to grow
- ❖ Environmental factors

Primary and secondary treatment of water ?

Primary Treatment: 1= Mechanical process 2= Screening 3= Grinding 4= Grit Chamber (heavy particles - “grit”) 5= Sedimentation Tank (suspended solids - “sludge”) 6= Chlorination of effluent

Secondary Treatment: 1= Biological process 2= Uses bacteria to remove organic matter dissolved in water (ODW) and lower BOD

Types 1= Trickling Filter 2= Activated Sludge

What are monogamous, polygynous and polyandrous.

1= Monogamous (single male pair with single female) 2= Polygynous (male mates more than one female) 3= Polyandrous (female mates more than one male).

What is parapatric speciation:

Parapatric speciation is the type of speciation when the species are separated due to an extreme change in their habitat when they enter a new habitat. Example: Grass *Agrostis tenuis*.

Orders of class reptilian:

Squamata, Testudines, Crocodilia and Rhynchocephalia

Define Light pollution

Light pollution, or artificial light at night, is the excessive or poor use of artificial outdoor light, and it disrupts the natural patterns of wildlife, contributes to the increase in carbon dioxide (CO₂) in the atmosphere, disrupts human sleep, and obscures the stars in the night sky.

Theory of “inheritance of acquired characteristics or soft inheritance”?

Jean Baptiste Lamarck (1744–1829) was a distinguished French zoologist. His contributions to zoology include important studies of animal classification. Lamarck, however, is remembered more for a theory of how change occurs. He believed that species are not constant and that existing species were derived from preexisting species. Lamarck's rather elaborate explanation for evolutionary change involved a theory that was widely accepted in the early 1800s called the theory of inheritance of acquired characteristics.

Function of peroxisomes:

Peroxisomes are organelles that sequester diverse oxidative reactions and play important roles in metabolism, reactive oxygen species detoxification, and signaling.

Orders of amphibians:

The three modern orders of amphibians are Anura (the frogs), Urodela (the salamanders), and Apoda (the caecilians).

Write down the coxal gland and Malpighian tubules?

The coxal gland is a gland found in some arthropods, for collecting and excreting urine.

Malpighian tubules (MTs) are the main osmoregulatory and excretory organs of insects and are considered analogous to the nephridia or kidneys, the corresponding organs of annelids and the vertebrates.

Nucleus made up of:

The nucleus is composed of various structures namely nuclear envelope, nucleoplasm or nucleolus, nuclear matrix, chromatin and nucleolus.

waste water treatment ?

Wastewater treatment is a process used to remove contaminants from wastewater and convert it into an effluent that can be returned to the water cycle. Once returned to the water cycle, the effluent creates an acceptable impact on the environment or is reused for various purposes (called water reclamation).[1] The treatment process takes place in a wastewater treatment plant. There are several kinds of wastewater which are treated at the appropriate type of wastewater treatment plant. For domestic wastewater (also called municipal wastewater or sewage), the treatment plant is called a sewage treatment plant. For industrial wastewater, treatment either takes place in a separate industrial wastewater treatment plant, or in a sewage treatment plant (usually after some form of pre-treatment). Further types of wastewater treatment plants include agricultural wastewater treatment plants and leachate treatment plants.